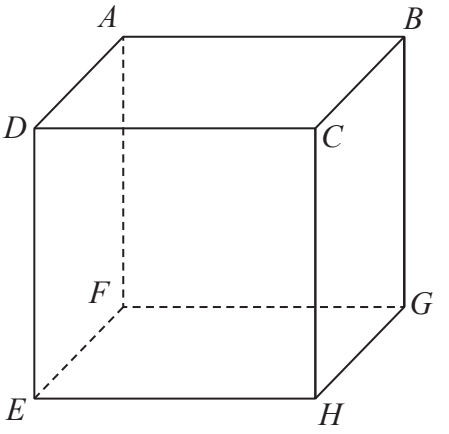
18 The diagram shows a cube.



AH = 11.3 cm correct to the nearest mm.

Calculate the lower bound for the length of an edge of the cube.
You must show all your working.

AH is the space diagonal of the cube.

Lower bound AH = 11.25 cm (11.3 cm to nearest mm).

For edge a: $AH^2 = a^2 + a^2 + a^2 = 3a^2$.

Lower bound $a = 11.25/\sqrt{3} \approx 6.49519$ cm.

4 marks total
AH lower bound 11.25
Space diagonal √3a
a=11.25/√3
≈6.495 cm

..... cm

(Total for Question 18 is 4 marks)

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